

Jin Rhee

jin.rhee@sjc.ox.ac.uk · +44 7442639381

EDUCATION

MEng in Engineering Science, St John's College, University of Oxford

Oct 2022 – June 2026 (expected)

Examinations:

Prelims: Pass (weighted avg. 78.22)

Part A: First (weighted avg. 77.11)

Scholarship:

\$55,000 per annum scholarship awarded by Kwanjeong Educational Foundation [[info](#)]

Korean Minjok Leadership Academy

Mar 2019 – Feb 2022

Examinations:

5s in ten AP subjects: Calculus BC, Chemistry, English Language, English Literature, Microeconomics, Macroeconomics, Physics (1, 2, C: Mechanics, C: Electricity and Magnetism)

Relevant courses:

Calculus, Vector Calculus, Linear Algebra, Differential Equations, Chemistry, Physics, Programming (C++), Applied Mathematics Project, English Literature

RELEVANT EXPERIENCE

Visiting Student Researcher, Urban Robotics Lab @ KAIST

Daejeon, Korea | Jul 2024 – Oct 2024

- Ran and compared trajectory error of robot odometry algorithms (FAST-LIO, LIO-SAM, and LeGo-LOAM)
- Implemented motion blur filter and open-source TSDF integration to improve quality of dense colour mesh generation from LiDAR-camera data fusion
- Refactored and improved real-time viability of room segmentation algorithm from scene graph optimized SLAM package for use in room-wise low-dynamic object detection
- Ground-truth labeling for ~2800 frames spanning 6 regions in point-wise dynamic object segmentation dataset available at [[HeLiMOS Dataset](#)]

Student Intern, Lightricity Ltd.

Oxford, England | Mar 2024 – Mar 2024

- Developed embedded C code for in-house ultra-low power indoor PV cell demo; established I2C communication with EPD display and lux sensor; implemented light-dependent sleep time adjustment on Epson 16-bit MCU
- Student micro-internship secured via Oxford University Careers Service; presented progress via presentation and handed bit register documentation

Mechatronics Engineer Intern, Bear Robotics

Seoul, Korea | Jun 2023 – Sep 2023

- Directly contributed to prototype development of gantry system for food bin placement; closely collaborated within small tight-knit team
- Built production code for simultaneous step-motor control; assessed mechanical repeatability and precision of gantry; optimized trajectory planning for speed and stability
- Prototyped clamps to isolate effect of rocker-bogie mechanism in robot base; experimentally confirmed previously unknown static tip-over condition of company's main robot model

Member, Team ORion @ Oxford Robotics Institute

Oxford, England | Jan 2023 – Apr 2023

- Developed pipeline comparatively testing speech recognition algorithms (Google Speech-to-Text API, Vosk Offline Speech Recognition API) for use on Toyota HSR (Human Service Robot)

PROJECTS / COURSEWORK

Project co-lead, QronkOxford @ OxRAM

Project, Oct 2022 – current

- Led undergraduate team to create low-cost 3D-printed 12 DOF quadruped robot; EUIF grant by Department of Engineering Science (Oxford); project available at [[GitHub](#)]
- Led motor control and power supply design; led tutorial sessions on Raspberry Pi, Ubuntu, Git, ROS, motor control basics, etc.; general reference for troubleshooting and handled purchase orders

AI/ML with Python

Coursework, June 2024

- Hands-on introduction to machine learning and mathematical fundamentals given in practicals and lectures
- Worked through regression/classification, optimization methods, and neural network problems in PyTorch on Jupyter notebook

OxCrest, Hack Cambridge Spyder Lite, Participant

Project, Mar 2023

- Created prompt-based college crest generator to overlay on existing photographs; available at [[GitHub](#)]
- Tested OpenCV Harris corner detection to locate crest location on image

X-Y Plotter Challenge, Oxford Robotics Institute, Participant

Project, Oct 2023

- Created 3D-printed 2-axis plotter involving CAD design, electronics testing, and firmware coding
- Wrote G-Code generator code to integrate with plotter control firmware

Development of 2-axis Self-Stabilizing cup holder

Project, Oct 2021

- Developed 2-axis cup-holder and examined PID tuning schemes for implementation using Arduino
- Presented poster at KIICE annual academic conference; paper available in [[Korean](#)]

ADDITIONAL SKILLS & INFO

Programming:

- C++, embedded C, Python, MATLAB

Proficient with:

- Ubuntu Linux (x86_64, aarch64), Arduino, Raspberry Pi

Experience with:

- CAD (Solidworks, Fusion360), ECAD (Eagle CAD), MCUXpresso
- LaTeX, 3D printing, rapid prototyping
- ROS (rospy, roscpp, rclcpp), Boost, PCL, Open3D, VoxBlox, GTSAM, g2o, evo,